

eBPF enhances IDS in the following ways.

Deep packet inspection (DPI): With eBPF, Snort and Suricata are able to inspect packets without expensive context switches.

System call monitoring: Traps malicious behaviour at the process level.

Performance efficiency: Unlike legacy packet sniffing, eBPF executes

within the kernel with reduced overhead.

Enhancing open source IDS with machine learning

While rule-based IDS products like Snort and Suricata work well, they must be updated with rules constantly to keep pace with evolving threats. Machine learning enhances these IDS products by identifying anomalies and novel attack

patterns. AI-based anomaly detection models scan behavioural patterns, flagging suspicious activity in real time.

Models for anomaly detection are:

Supervised learning: Uses labelled datasets to distinguish between normal and malicious activity.

Unsupervised learning: Finds anomalies by detecting the variance from usual behaviour.

Deep learning: Neural networks can learn to identify complex attack patterns in high volumes of data.

Here's a sample Python code for anomaly detection in IDS:

```
from sklearn.ensemble import
IsolationForest
import numpy as np

# Sample network traffic data
data = np.random.rand(100, 5) # 100
samples, 5 features
model = IsolationForest(contaminati
on=0.05)
model.fit(data)
anomalies = model.predict(data)
print("Detected Anomalies:",
sum(anomalies == -1))
```

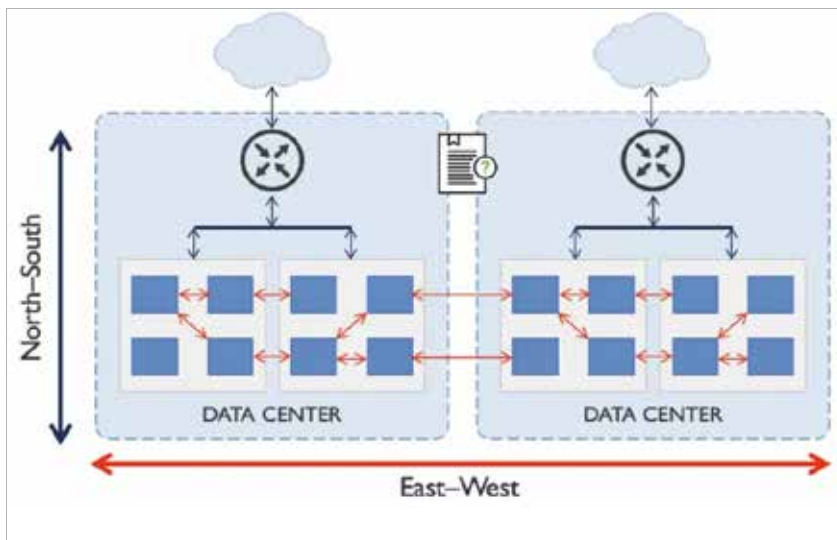


Figure 1: East-West vs North-South traffic in Kubernetes security

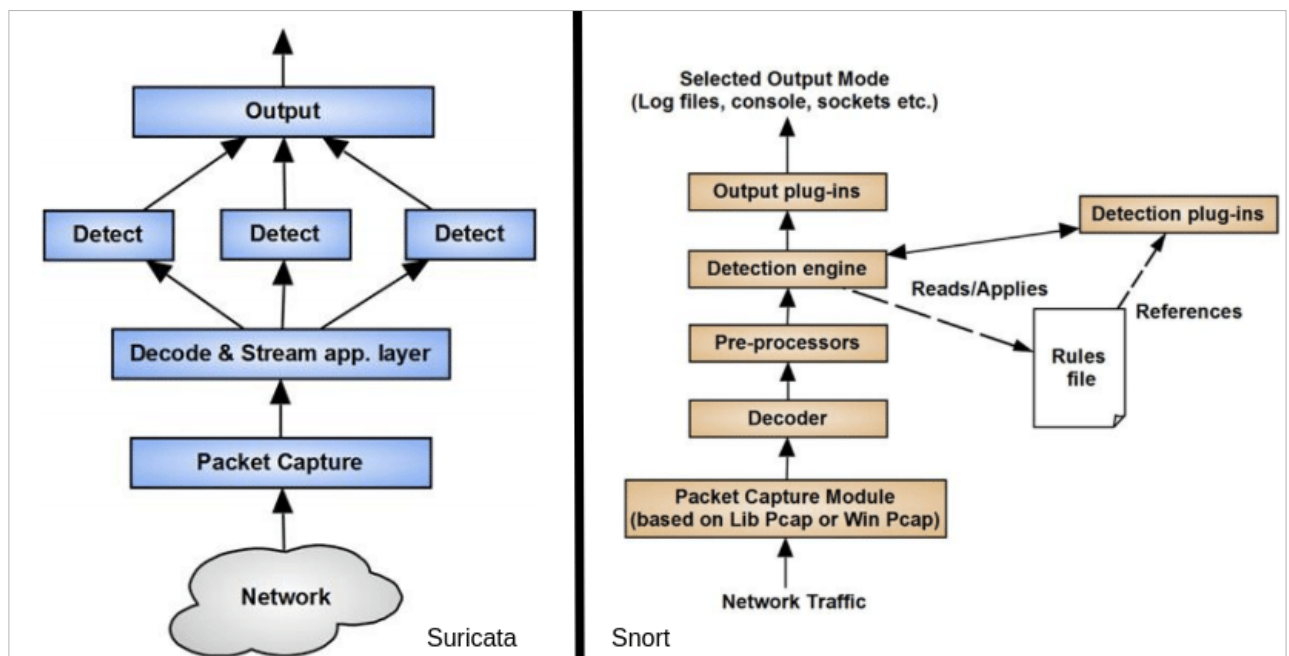


Figure 2: Packet processing workflows in Snort and Suricata architecture